$$= \frac{1}{2}\int\left[\frac{1}{z^2} - \frac{1}{z^3}\right]dz$$

$$= \frac{1}{2}\left[-\frac{1}{z} + \frac{1}{2z^2}\right] + c$$

$$= \frac{1}{4\left(x^2+1\right)^2} - \frac{1}{2\left(x^2+1\right)} + c$$

Example 2.34

Evaluate $\int \frac{dx}{x\left(x^3+1\right)}$

Solution:

Take $z = x^3$	By partial fractions,
$\therefore\ dz = 3x^2\,dx$	$\frac{1}{z(z+1)} = \frac{A}{z} + \frac{B}{z+1}$
$\Rightarrow \frac{dz}{3} = x^2\,dx$	$\Rightarrow \frac{1}{z(z+1)} = \frac{1}{z} - \frac{1}{z+1}$

$$\int \frac{dx}{x\left(x^3+1\right)} = \int \frac{x^2}{x^3\left(x^3+1\right)}dx$$

$$= \frac{1}{3}\int \frac{1}{z(z+1)}dz$$

$$= \frac{1}{3}\int\left[\frac{1}{z} - \frac{1}{z+1}\right]dz$$

$$= \frac{1}{3}\left[\log|z| - \log|z+1|\right] + c$$

$$= \frac{1}{3}\log\left|\frac{z}{z+1}\right| + c$$

$$= \frac{1}{3}\log\left|\frac{x^3}{x^3+1}\right| + c$$

Example 2.35

Evaluate $\int x^3 e^{x^2}\,dx$

Solution:

Take $z = x^2$

$\therefore\ dz = 2x\,dx$

$\Rightarrow \frac{dz}{2} = x\,dx$

$$\int x^3 e^{x^2}\,dx = \int x^2 e^{x^2}\,(x\,dx)$$

$$= \frac{1}{2}\int z\,e^z\,dz$$

$$= \frac{1}{2}\left[e^z\,(z-1)\right] + c \quad \text{[By applying Integration by parts]}$$

$$= \frac{1}{2}\left[e^{x^2}\,(x^2-1)\right] + c$$

Example 2.36

Evaluate $\int e^x\left(x^2+2x\right)dx$

Solution:

Take $f(x) = x^2$

$\therefore\ f'(x) = 2x$

$$\int e^x\left(x^2+2x\right)dx = \int e^x\left[f(x) + f'(x)\right]dx$$

$$= e^x\,f(x) + c$$

$$= e^x\,x^2 + c$$

Example 2.37

Evaluate $\int \frac{xe^x}{(1+x)^2}dx$

Solution:

By partial fractions,	Take
$\frac{x}{(1+x)^2} = \frac{A}{1+x} + \frac{B}{(1+x)^2}$	$f(x) = \frac{1}{1+x}$
$\Rightarrow \frac{x}{(1+x)^2} = \frac{1}{1+x} + \frac{-1}{(1+x)^2}$	$f'(x) = \frac{-1}{(1+x)^2}$

$$\int \frac{xe^x}{(1+x)^2}dx$$

$$= \int e^x\left[\frac{1}{1+x} + \frac{-1}{(1+x)^2}\right]dx$$

$$= \int e^x\left[f(x) + f'(x)\right]dx$$

$$= e^x\,f(x) + c$$

$$= \frac{e^x}{1+x} + c$$

Example 2.38

Evaluate $\int e^{2x}\left[\frac{2x-1}{4x^2}\right]dx$

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